Project Final Report

MFB2102 Engineering Team Project II

Title	RainGuard: Smart Retractable Roof System	**Group Number**	18
Team Name	The Inventor		

Team Members

The team members are as follows:

No.	Name	ID	Dept.
1	Devaline Zheyrra Jeanesya	25005916	CHE
2	Aisar Bin Abdul Rahman	22010582	CHE
3	Muhammad Rieqhmal Mukhreez Bin Rozmi	24003469	COE
4	Idriss Rama Salim	24001751	EE
5	Zaim Bazli Bin Mohd Arifin	22010438	ME
6	Nur Faten Syakirah Binti Ahmad Farid	22010059	PE

The leader of the team is **Mr Muhammad Rieqhmal Mukhreez Bin Rozmi**

Coach

AP Dr Norhayati binti Mellon

Problem Statement & Context (Discover Phase)

Demonstrate understanding of the real industry problem and the larger context in which the problem exists.

Introduction

In recent years, many individuals have become increasingly occupied with work, studies and household responsibilities, making it difficult to manage time efficiently. At the same time, weather patterns have become more unpredictable compared to previous decades. What were once clearly defined as rainy and dry seasons has now shifted into irregular and sudden changes, often causing inconvenience in daily routines and contributing to reduced productivity.

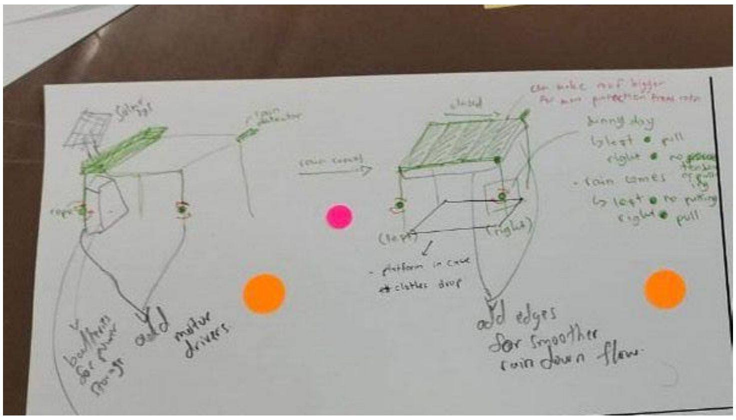
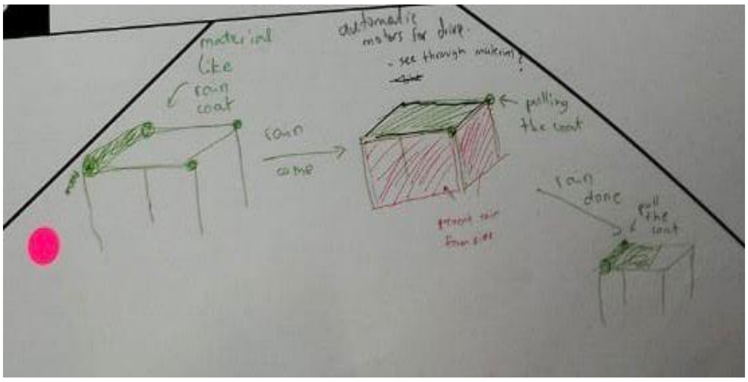
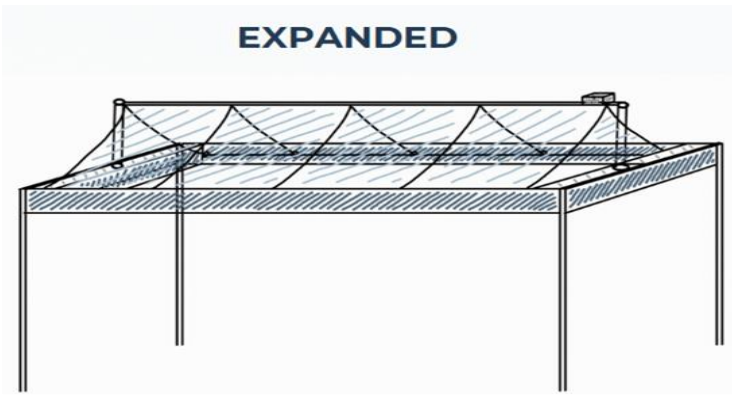
One common problem faced by many households, particularly students living in semi-outdoor residential environments, is the difficulty of drying clothes during unpredictable weather. Sudden rainfall can cause clothes to become wet again, forcing users to rewash or re-dry their laundry, which waste time, water and effort.

To address this challenge, RainGuard is proposed as an automated protective system designed to protect clothes from unexpected rain. The system uses a rain detection sensor that trigger a retractable roof to cover the clothes automatically when rainfall

	is detected. By providing a simple solution, RainGuard aims to reduce inconvenience and support more efficient routines.
Background Research	Unpredictable weather conditions are becoming increasingly common in many tropical regions, including Malaysia. Sudden rainfall often occurs throughout the year, making it difficult for households to plan outdoor activities such as drying clothes. In many residential areas, clothes are typically dried in semi-outdoor spaces such as corridors or balconies where sunlight and airflow are available. However, these areas offer limited protection from unexpected rain. At Universiti Teknologi PETRONAS (UTP), this issue is also experienced by many students. In the residential villages, not all drying rack areas are covered. As a result, clothes that are left outside to dry can easily become wet again during sudden rainfall. Currently, several solutions exist to address similar problems. Electric clothes dryers allow clothes to dry quickly regardless of weather conditions, but they consume too much electricity. Some households install retractable awnings or covered drying areas, but these systems are often expensive, require permanent installation and are designed for larger residential buildings rather than small scale personal use. Based on the survey conducted among UTP students, 77% of respondents reported experiencing clothes getting wet again because of rain, while 79% are interested or very interested with the concept of an automatic retractable roof. These findings highlight a clear need for a simple, affordable and automated solution that can help students managing their laundry efficiently.
Stakeholder Identification	1. Keep Satisfied (High Influence, Low Interest) • **UTP Facilities Management / HSE Department** These stakeholders have the authority to approve or restrict any installation within hostel corridors based on safety regulations. Although they are not directly concerned with the daily use of the drying system, it is important to ensure that the design complies with safety standards and does not pose any hazards.

	2. Manage Closely (High Influence, High Interest) • **Residential Village (RV) Supervisors** RV Supervisors are responsible for maintaining the condition and order of hostel environments. They are highly concerned about corridor usage, clutter and student discipline. Since they have the authority to remove unapproved items and a strong interest in maintaining order, they must be actively consulted and involved throughout the project. 3. Monitor (low Influence, Low Interest) • **Campus Visitors / Lecturers** These stakeholders have minimal interaction with the system, as they do not reside in the hostels or use the drying facilities. Their involvement is limited to visual exposure. Therefore, minimal management effort is required. 4. Keep Informed (Low Influence, High Interest) • **Hostel Residents (Students)** These are our primary users. They have a high level of interest as the solution directly addresses their daily challenges with drying clothes. However, they have limited authority in decision-making regarding installations. Therefore, they should be kept informed about how the system works and its proper usage.
Personas	Amir is a second year UTP student living in the hostel. Like many students, he constantly balances classes, labs, meetings and assignment deadlines. Laundry is a task he oft6en find inconvenient. The shared drying areas in the hostels are limited, and his clothes are usually hung in semi-outdoor areas. Due to sudden rain or high humidity, his clothes sometimes become wet again, requiring extra time and effort to rewash or re-dry them. Amir prefers a simple and practical solution that does not require complicated setup or constant monitoring. He needs a system that can protect his clothes automatically, especially during busy academic periods when he cannot always check the weather.

	The proposed *RainGuard: Smart Retractable Roof System* is designed to address these challenges. With its rain-detection feature and clamp-based installation, the system can automatically deploy a protective roof when rain is detected without requiring permanent installation. This makes it suitable for hostel environments where flexibility and ease of use are important. For Amir, this solution represents a more convenient and efficient way to manage his daily routine, reducing the hassle of laundry and allowing him to focus on his academic and personal activities.
Problem Definition (Define Phase) *Narrow down and clearly articulate the specific challenge the team are addressing.*	
Opportunity Statement	The core challenge addressed by this project is the lack of a simple, automated, and non-permanent way for students in semi-outdoor hostel environments to protect their laundry from unpredictable weather. How might we design an affordable, automated protective system for UTP hostel students that prevents laundry from getting wet during sudden rain while remaining easy to install and remove?
Key Criteria for Success	To determine the success of the RainGuard system, the following technical and user-centred criteria must be met: • Automated Detection: The system must accurately detect water presence in real-time using a dedicated rain sensor. • Responsiveness: The retractable roof must deploy immediately upon contact with water and retract once the sensor is dry. • Non-Permanent Installation: The system must utilize a clamp-based mechanism to allow for easy attachment to existing drying racks without requiring permanent structural modifications. • Reliability & Durability: The structure must be lightweight enough for motor operation but durable enough to withstand outdoor moisture using waterproof materials. • User Satisfaction: The solution must specifically address student needs for time efficiency and reduced laundry re-washing.

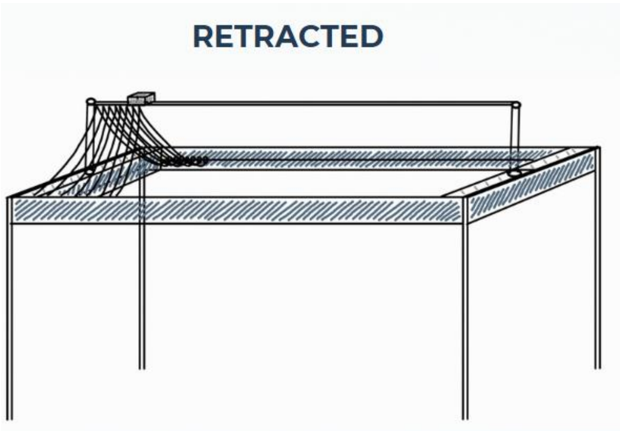
Ideation & Concept Development (Develop Phase) *Explore multiple potential solutions, demonstrating creative problem-solving and feasibility analysis.*	
Collaborative Sketching	The team engaged in a C-Sketch (Collaborative Sketching) process involving all six members to explore various approaches to automated weather protection. This iterative process allowed the team to refine the mechanical movement of the roof and the placement of electronic components. The top three ideas that the team voted for are:   

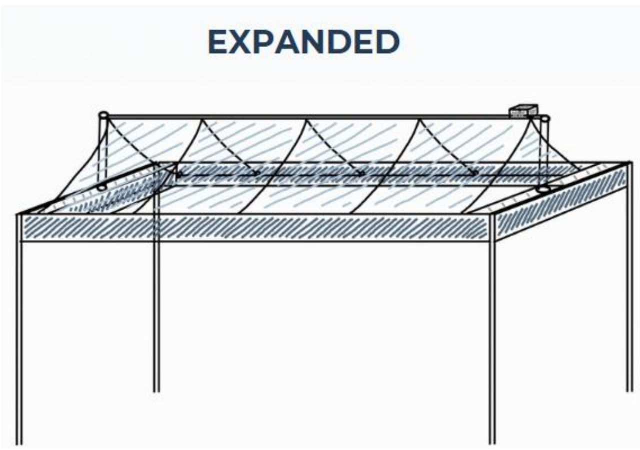
Real-Win-Worth	The Smart Retractable Roof System was selected as the final concept based on its high feasibility and alignment with stakeholder needs: • Feasibility: The design uses accessible, cost-effective components such as the Arduino Uno, the SN-RAIN-MOD sensor, and a high-torque JGB37-3530 motor. Using parachute material ensures the roof is both waterproof and lightweight, minimizing the load on the motor driver. • Alignment with Stakeholder Needs: * Students (Amir Persona): Provides the automation needed for students with tight schedules who cannot manually monitor the weather. • RV Supervisors: The clamp-based, non-permanent design ensures the system complies with hostel regulations regarding clutter and unauthorized permanent installations. • Economic Worth: With a production cost of RM161.09, the system remains affordable for personal use while offering a significant advantage over expensive permanent awnings or high-energy-consuming electric dryers.
Solution & Development (Develop Phase) *Outline the project's solution and development.*	
Detailed Solution Overview	We installed a smart retractable roof system to the existing drying rack in UTP. The roof is equipped with rain drop sensor module that detects water presence and automatically send a signal to the motor to spin so that the roof can be deployed. When the rain stops, sensor will no longer detect the water presence and then make the roof retract back to its original place. The roof system's size is also adjustable, suitable for wide range of rack's size. The installation of the roof is easy enough to let the user install it to another rack. Functional specification: • Sensors to detect rain in real-time. • A powerful motor that can pull • Adjustable roof size • Rechargeable batteries

System Technical Specifications:

Sensor Type	SN-RAIN-MOD sensor used to accurately measure water presence.
Motor	JGB37-3530 111rpm motor that have 7kg.cm rated torque and a stall torque of up to 24kg.cm
Power Source	3x 12000 mAh rechargeable lithium batteries.
Material	Roof using a parachute material to ensure waterproof and lightweight are in check.
Microcontroller	Arduino Uno for processing sensor data and managing DC Motor operations using C++ language.

The wires are properly bundled, secured, and routed to minimize the risk of chafing, abrasion, or short circuits. Weather-resistant enclosure was used and sealing material to protect electrical components from moisture.





Project Milestones

We have scheduled our weekly activities as below:

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												

Legend:

1. Conducted meeting among members to discuss items needed for purchase.
2. Purchased required items.
3. Do some minor tweaks and adjustments to the product.
4. Finalizing the product design.
5. Start working on the electronics part of the product.
6. Fill up the claim form.
7. Building the mechanical part in Garage 21.
8. Writing final report.
9. Presenting the finished product to advisor before ETPx.
10. Presenting the product during ETPx.

Roles and Responsibilities

The role for each member was assigned and agreed as below:

No.	Name	Role
1	Muhammad Rieqhmal Mukhreez bin Rozmi	Roof behaviour programming (Arduino). Also act as a leader.
2	Idriss Rama Salim	Electrical and electronics circuit building.
3	Zaim Bazli bin Mohd Arifin	Lead the product’s mechanical design and build.
4	Devaline Zheyrra Jeanesya	Roof material and environmental impact assessment. Also act as a treasurer.
5	Nur Faten Syakirah binti Ahmad Farid	Roof placement strategy and structural integrity.
6	Aisar bin Abdul Rahman	Designing and building the mechanical part of the product.

	List down 3 principles (ground rules) that the team subscribed to for this project. 1. Always keep in touch with each other, whether it is ideas, challenges or updates. Keep an open mind to new ideas. 2. Take accountability of our task. Always meet the deadline and give 100% commitment. 3. Always find a problem before it appears. There is always a room for an improvement.
Project Implementation (Deliver Phase) *Discuss the project development, testing, iteration and final product.*	
Prototype Development	The prototype development began with assembling the electrical components of the RainGurad system. The rain sensor module, Arduino Uno microcontroller, motor driver and rechargeable battery were first connected and tested to ensure proper signal transmission and motor response. The system was programmed so that the motor would activate when the sensor detect water. After the successful testing of electrical system, the team proceed to construct the mechanical structure. A retractable roof mechanism was developed using lightweight materials with waterproof parachute fabric to ensure durability while minimizing load on the motor. A clamp-based installation system was designed and attach to the structure to allow the product to be easily installed onto existing drying racks without requiring permanent modification. After both the electrical and mechanical components were completed, the system was combined into a single unit. The rain sensor was positioned in an exposed area to detect rainwater while the motor was connected to the roof mechanism to enable automatic retraction of the roof. The wiring was carefully arranged and secured to ensure safe operation. Several challenges were encountered during the development process. One of the main challenges was ensuring the motor had sufficient torque to retract the roof while maintaining smooth operation. Additionally, achieving a stable yet lightweight roof structure that could be efficiently operated by the motor required multiple

	adjustments. These challenges were address through iterative testing and reinforcement of the mechanical structure.
Testing and Iterations	Once the prototype was completed, testing was conducted to evaluate its functionality. A rain detection test was carried out by applying water directly onto the rain sensor. The rain sensor successfully detected the presence of water and triggered the motor to deploy the retractable roof. When the sensor no longer detected water, the roof retracted back to its original position. Additionally, the prototype was demonstrated to several students who regularly use the drying racks at the hostel. Feedback from the students indicated that the system was useful as it reduces the need to monitor weather conditions. Some minor improvements were suggested including enhancing the smoothness of the roof movement.
Final Design	The final design of RainGuard consists of a clamp-based retractable roof system with a rain sensor, Arduino Uno microcontroller, DC motor and rechargeable battery. The system automatically detects rain and deploys the roof to protect clothes and retracts back when the rain stops. The usage of clamp allows easy installation onto the existing drying rack at UTP residential villages without requiring permanent installation. The final design of the product provides a practical and user-friendly solution to protect clothes from unexpected rain.
Result and Impact *Deliberate the result gained from the solution implementation.*	
Results	The effectiveness of the RainGuard system was evaluated based on its ability to detect rain and automatically deploy the protective roof. During testing, the rain sensor successfully detected the presence of water and triggered the motor to deploy the retractable roof. The system was able to respond immediately upon contact with water and retract the roof when the sensor was dry. The system demonstrates consistent performance during testing which show that the product is reliable. The results also show the potential of the system to be scaled to UTP residential village areas.

<table>
<tr>
<td>User Feedback</td>
<td>Students, the primary users of the RainGuard system, expressed satisfaction with its functionality. Students found the automatic rain detection feature to be highly useful as it reduces the need to constantly monitor weather conditions. In addition, users appreciated the clamp-based installation, as it allowed the system to be easily attached to existing drying racks without requiring permanent modification. Overall, the system was perceived as useful in improving daily laundry routines in hostel environments.</td>
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<tr>
<td>Environmental, social, and governance (ESG) Impact</td>
<td>
<table>
<tr><th>Environmental</th><th>Social</th><th>Governance</th></tr>
<tr>
<td>Reduces water and electricity usage by preventing clothes from getting wet and having to be rewashed.
Decrease energy usage by encouraging traditional drying instead of electric dryer.</td>
<td>Prevent inconvenience caused by sudden rain, especially for students with tight schedules.
Save times as users do not have to rewash clothes that have been soaked with rain.</td>
<td>Enhances safety by ensuring proper wiring and using waterproof components.
Provide usage guidelines for students to ensure proper operation.</td>
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<tr>
<td colspan="2">Challenges and Lessons Learned
Outline the ups and downs while executing this project.</td>
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<tr>
<td>Technical Challenges</td>
<td>There were several issues encountered during the prototype development process. The first issue is the sudden need to change the design. The previous design has a simpler technical design. However, it could not build for mass. In other words, the cover area is much smaller and not applicable for several drying rack types. Therefore, we re-evaluate and improved the design before the industry touchpoint session.</td>
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	During the industry touchpoint, we were suggested to improve the design by making the design more applicable to all drying rack types. We introduced a more flexible features in our smart roof system by making it adjustable in both width and length. The improved design needs different types of motor, which required several times of brainstorming to determine the suitable motor work design. After finalizing the motor design, additional challenges were encountered during the construction of the smart roof system. The adjustable features have more complicated technical detail, especially since the materials are also limited. However, these challenges were resolved through further discussion and adjustments and the prototype was successfully completed.
Teamwork and Collaboration	Our team has a great communication improvement, it did not take a long time to collaborate and discuss intensely, both in formal and nonformal. Each member naturally has a role whether in technical or social which made the team bonding grew faster. When challenges appeared during the project, we discussed them together and tried to find the best solution. We always listen, respect, and support the ideas which could improve and advance the project. As a result, our tasks were finished more effectively with a smooth workflow. In facing issues and challenges, we were able to adapt quickly to changes the ideas which helped the team stay flexible and continuous to the progress. Each member also showed responsibility for the assigned tasks, which made the collaboration more organized and reduced misunderstandings during the project. This effective collaboration can be achieved easily due to the positive team environment. It made everyone feel comfortable to sharing ideas and opinions, which helped us develop better ideas together. Overall, the teamwork and collaboration played an important role in achieving the project goals successfully.

Budget and Resource Allocation *Tabled the expenses for this project.*	
Expenses breakdown	The expenses details are as follow:

Item	Price
AA Battery Holder	MYR 1.79
MG995-180Deg (Metal) Servo Motor	MYR 16.90
Water Raindrops Weather Rain Sensor Module - Analog & Digital for Arduino	MYR 6.50
AA4 Carbon Battery	MYR 5.90
4 mm Starter Rope	MYR 2.40
PVC	MYR 6.00
Cable Ties	MYR 3.50
Metal Roller	MYR 3.00
Rope	MYR 1.00
Raincoat	MYR 3.30
Window Washer	MYR 15.00
Rechargeable 18650 Lithium Battery 20000mAh Flat Top	MYR 8.37
Rechargeable 18650 Lithium Battery 20000mAh 2 Slot Charger (+shipping fee)	MYR 7.81
Lithium 18650 battery holder with coil spring contact	MYR 4.50
DC JGB37-12V-111RPM	MYR 39.90
DC Motor Driver Smart Current Sensor Drive (+shipping fee)	MYR 19.42
DC-DC Adjustable Converter Module	MYR 3.00
(4Pcs) Value Pack Alkaline Battery	MYR 12.80
Total	**MYR 161.09**

The budget given by the ETP Committee is RM500 which can cover all the expenses for project's items needed. Additionally, to anticipate any failure or emergency needs, a 10% of budget is aside for emergency fund.

	<table><tr><td>Budget</td><td>MYR 500.00</td></tr><tr><td>Total Expenses</td><td>MYR 161.09</td></tr><tr><td>Emergency Fund</td><td>MYR 50.00</td></tr><tr><td>Remaining Funds</td><td>MYR 288.91</td></tr></table> Therefore, the cost of making the Rainguard is RM161.09, and the remaining funds for future expenses available is RM288.91.
Commercial Market	**Given:** Cost = RM161.09 Profit Margin = 35% **Profit Calculation:** Profit = $\frac{\text{Profit Margin (\%)}}{100\%} \times \text{Cost}$ = $\frac{35\%}{100\%} \times \text{RM161.09}$ = RM 56.38 **Selling Price** Selling Price = Cost + Profit = RM161.09 + RM56.38 = RM217.47 = RM217.50 The profit margin of 35% is considered as reasonable value since it is a prototype-based products. This consideration is due to high production cost unlike mass production products which their margins can be lowered. Mass production, such as industrial, can get mass of raw material with lower cost. Meanwhile prototype-based product can only be constructed using commercial stuffs. In conclusion, the profit obtained per product is RM56.38 so the selling price will be RM217.50, round up of RM217.47.

Sustainability and Future Recommendations *Ensuring Long-Term Sustainability and Development.*	
Scalability	The Rainguard smart roof system has a potential to be scaled up for deployment in dormitories/villages. This product has a potential in high demand since it also affordable and applicable.
Improvements and Future Work	Future iterations of the project can be focused on solar panel to replace the battery as the power source to increase the efficiency of energy and to support the SDGs. Additionally, it is highly recommended to incorporating app-based monitoring system featured current weather and weather forecast as a sign and announcement of the smart roof to be expanded. The team also suggests collaborating with manufactures to reduce the production cost which highly affect the selling price. More affordable products have higher potential of wider consumer.
Conclusion *Summarize the key points and outline the immediate next steps post-proposal submission.*	
Closing of the project	The Rainguard smart roof system successfully helped the target audience to avoid daily issue such as laundry. This is aligned with our goal to help students increase their productivity, not to overthink about household matters and focus on education instead.
Appendices	
References	Tan, X., Yang, L., He, Y., & Chen, J. (2023). Research on IoT-based smart clothes drying rack. In 2023 5th International Conference on Internet of Things, Automation and Artificial Intelligence (IoTAAI 2023) (pp. 275–281). ACM. https://doi.org/10.1145/3653081.3653127 Weather & Climate. (n.d.). Seri Iskandar (MY) average monthly precipitation – rainfall. Retrieved November 22, 2025, from https://weather-and-climate.com/average-monthly-precipitationRainfall,seri-iskandar-my,Malaysia webotricks. (2025, March 27). Rain Detection System Using Arduino. Arduino Project Hub. Retrieved November 22, 2025, from https://projecthub.arduino.cc/webotricks/rain-detection-system-usingarduino-aafbf5

Additional Visuals/ Diagrams

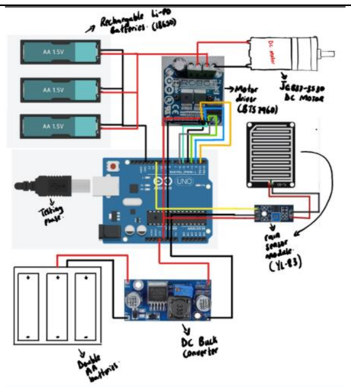


Figure 1. Circuit diagram

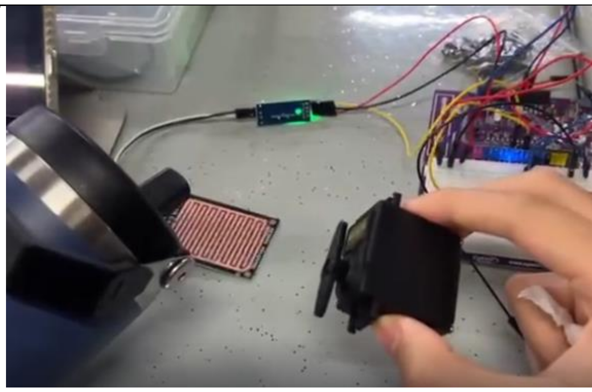


Figure 2. Core technical aspect test

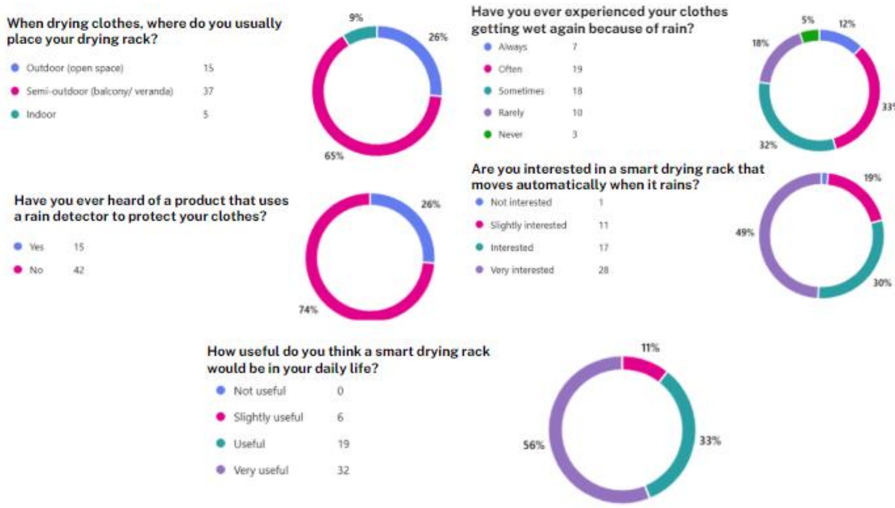


Figure 3. Survey result

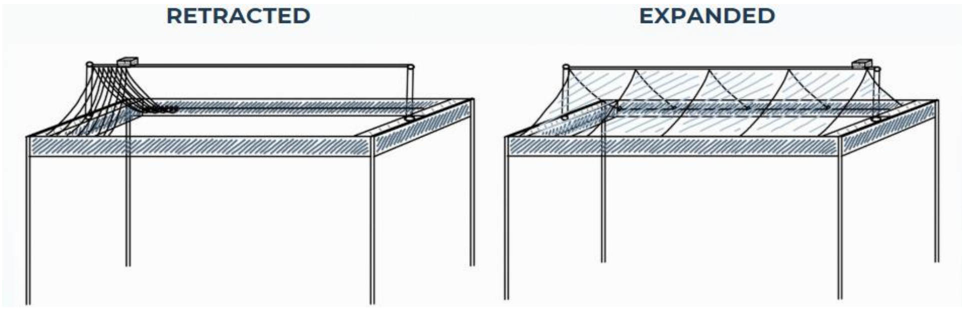
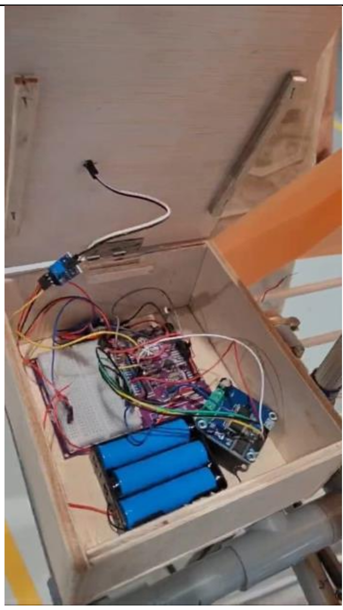


Figure 4. C-Sketch design

	 *Figure 5. Core technical box*  *Figure 6. Product applied to certain drying rack type*